

Metals & Minerals Supply Monitor

Summary analysis, data through Sep 2026 (window 2019-01 to 2026-09). 15 sites and 5 climate regions.

SYNTHETIC DEMO DATA: random test series generated to show the report layout. Nothing in this document is an observation.

15

sites monitored

8

commodity indices

2

climate / power regions
flagged

28

2-sigma signal moves, latest
month

Key takeaways

- Nickel: supply-activity index +1.8 (3-month average z) across 3 site(s), well above its seasonal norm. Largest site moves: IWIP Weda Bay +2.4; IMIP Morowali +2.3.
- Lithium: supply-activity index -0.8 (3-month average z) across 2 site(s), below its seasonal norm, down 0.9 vs. three months earlier. Largest site moves: Greenbushes -0.6; Salar de Atacama ponds -0.5.
- Palladium: supply-activity index +0.6 (3-month average z) across 1 site(s), above its seasonal norm, up 0.4 vs. three months earlier. Largest site moves: Norilsk complex +0.6.
- Gold: supply-activity index -0.5 (3-month average z) across 1 site(s), near its seasonal norm, down 0.5 vs. three months earlier. Largest site moves: Grasberg district -0.5.
- Yunnan hydro system: 3-month rainfall at 43% of normal (SPI-3-like -4.3, SEVERE DRY) to Aug 2026. Exposure: aluminum. Hydro-powered aluminium smelting; drought leads to power curtailment (cf. 2022-23).
- Upper Zambezi / Kariba catchment: 3-month rainfall at 80% of normal (SPI-3-like -1.2, DRY) to Aug 2026. Exposure: copper. Kariba hydro supplies the Zambian copperbelt; drought leads to load-shedding (cf. 2024).
- 28 signal reading(s) moved 2+ standard deviations in Sep 2026. Largest: IMIP Morowali surface change +4.4; Morenci pond / water area +4.0; IMIP Morowali tree cover -3.8.
- Fastest footprint growth relative to its own history: IWIP Weda Bay (Halmahera) (capacity index +4.4).
- Strongest price link: sichuan SPI-3 -> aluminum price ($r = +0.31$ at lag 2, signal leads price; $n = 91$). Best of 13 lags, so treat it as a hypothesis to test.

Commodity supply activity

Commodity	Utilization (3m z)	Change vs. 3m ago	Capacity (z)	Sites	Largest site moves (3m z)
Nickel	+1.8	+0.2	+4.0	3	IWIP Weda Bay +2.4; IMIP Morowali +2.3
Lithium	-0.8	-0.9	+2.2	2	Greenbushes -0.6; Salar de Atacama ponds -0.5
Palladium	+0.6	+0.4	-	1	Norilsk complex +0.6
Gold	-0.5	-0.5	+1.3	1	Grasberg district -0.5
Iron Ore	+0.2	+0.1	+1.5	2	Mt Whaleback / Newman +0.4; Port Hedland export hub +0.2
Copper	-0.2	+0.0	+2.1	8	Kamoa-Kakula +2.5; Cobre Panama pits + plant -1.6
Aluminum	+0.0	+0.1	+2.0	1	Port Kamsar bauxite terminal +0.0
Bauxite	+0.0	+0.1	+2.0	1	Port Kamsar bauxite terminal +0.0

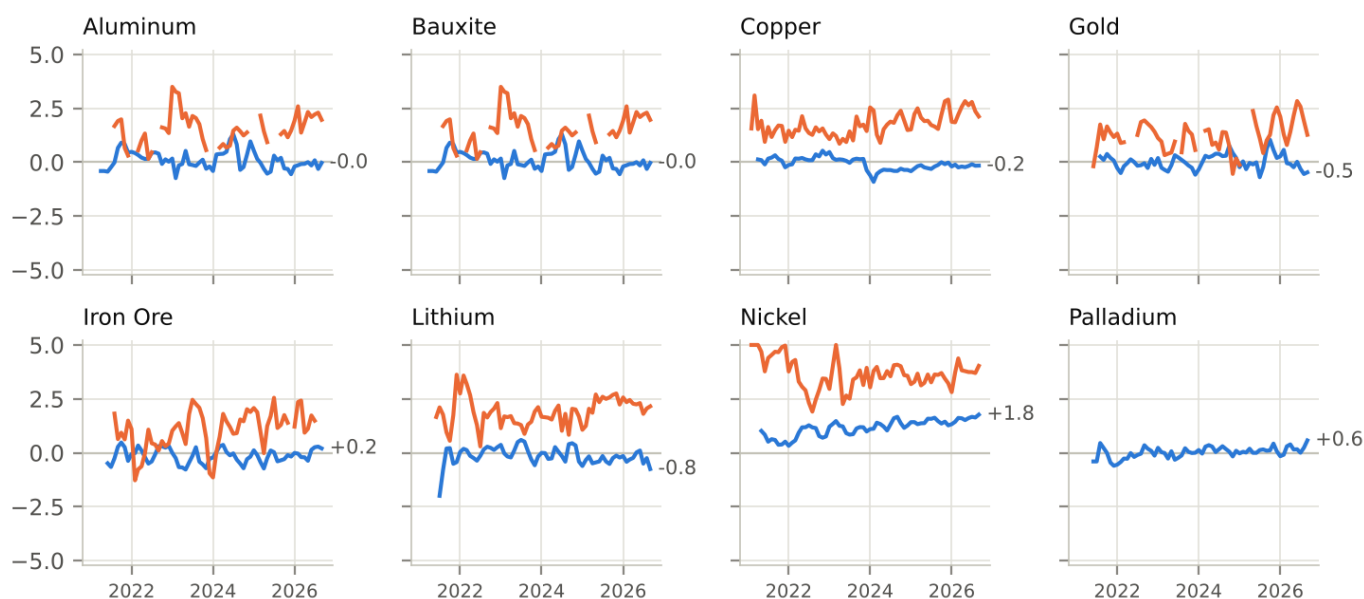
Cell shading: blue above, red below each series' own seasonal norm; the value is always printed. Supply is roughly capacity x utilization.

Commodity and site activity

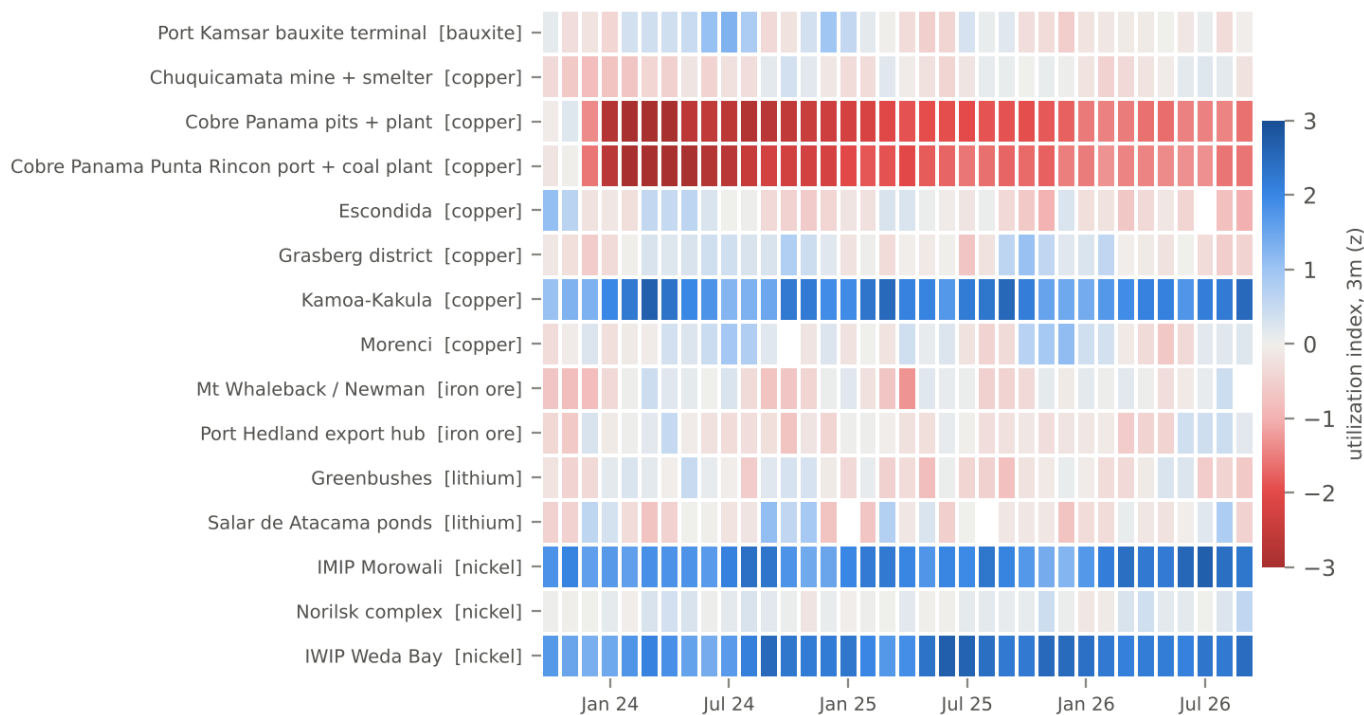
Utilization averages throughput proxies (SO_2/NO_2 enhancement, night lights, thermal hotspots, surface and SAR change). Capacity tracks footprint growth (bare/built area, pond area for brine). Both are z-scores against the site's own history up to that month.

Commodity supply-activity indices (z)

— Utilization (3m) — Capacity



Site utilization index, last 36 months (grey = no data / normal)



Climate and power-supply risk

Region	Exposure	Risk side	Month	3m rain % normal	SPI-3-like	Temp anom. (C)	Flag
Yunnan hydro system	aluminum	dry	Aug 2026	43%	-4.3	+0.3	SEVERE DRY
Sichuan hydro system	lithium, aluminum	dry	Aug 2026	103%	+0.2	+0.8	-
Upper Zambezi / Kariba catchment	copper	dry	Aug 2026	80%	-1.2	-0.1	DRY
Boke bauxite region	bauxite, aluminum	wet	Aug 2026	98%	-0.1	+1.1	-
Pilbara iron ore belt	iron ore	wet	Aug 2026	71%	-1.9	+0.5	-

Yunnan hydro system: Hydro-powered aluminium smelting; drought leads to power curtailment (cf. 2022-23).

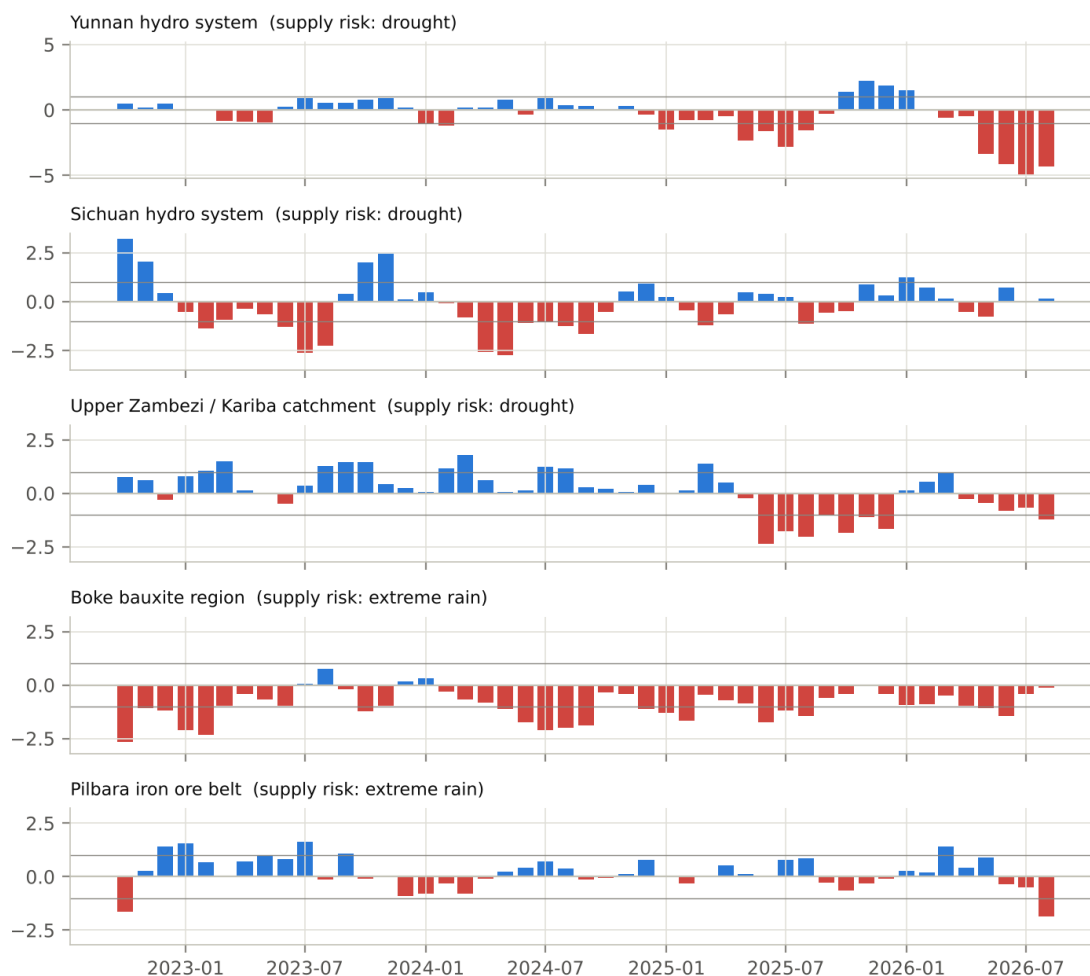
Sichuan hydro system: Hydro-heavy grid; heat plus drought leads to industrial power rationing (cf. Aug 2022).

Upper Zambezi / Kariba catchment: Kariba hydro supplies the Zambian copperbelt; drought leads to load-shedding (cf. 2024).

Boke bauxite region: An extreme wet season disrupts haulage and loading, so exports dip.

Pilbara iron ore belt: Cyclone and heavy-rain season (about Dec-Apr) causes rail and port outages.

3-month rainfall anomaly, SPI-3-like (blue wetter, red drier; lines at +/-1)



Site snapshot

Site	Commodities	Month	Util. (z)	Util. 3m	Capacity	Top driver
Escondida (BHP)	copper	Sep 2026	-0.9	-1.0	+1.4	night lights (-0.9)
Chuquicamata mine + smelter (Codelco)	copper	Sep 2026	-0.4	-0.2	+3.2	NO2 (-0.7)
Kamoa-Kakula (Ivanhoe / Zijin)	copper	Sep 2026	+2.3	+2.5	+3.3	night lights (+3.2)
Cobre Panama pits + plant (First Quantum)	copper	Sep 2026	-1.6	-1.6	+2.9	SAR change (-1.9)
Cobre Panama Punta Rincon port + coal plant	copper	Sep 2026	-1.4	-1.6	-	night lights (-1.8)
Grasberg district (Freeport)	copper, gold	Sep 2026	+0.0	-0.5	+1.3	night lights (+1.6)
Morenci (Freeport)	copper	Sep 2026	+0.2	+0.2	+1.0	night lights (+0.3)
Norilsk complex (Nornickel)	nickel, palladium, copper	Sep 2026	+1.3	+0.6	-	night lights (+2.1)
IMIP Morowali (Tsingshan et al.)	nickel	Sep 2026	+2.9	+2.3	+3.6	surface change (+4.4)
IWIP Weda Bay (Halmahera)	nickel	Sep 2026	+2.4	+2.4	+4.4	night lights (+3.3)
Greenbushes (Talison)	lithium	Sep 2026	-0.5	-0.6	+2.2	night lights (-0.7)
Salar de Atacama ponds (SQM / Albemarle)	lithium	Sep 2026	-1.7	-0.5	+2.1	surface change (-1.7)
Port Hedland export hub	iron ore	Sep 2026	+0.7	+0.2	+1.5	surface change (+0.9)
Mt Whaleback / Newman (BHP)	iron ore	Jul 2026	+0.4	+0.4	+1.5	night lights (+0.4)
Port Kamsar bauxite terminal (CBG)	bauxite, aluminum	Sep 2026	+0.3	+0.0	+2.0	SAR change (+1.0)

Recent anomalies (|z| of 2 or more, last 6 months)

Site	Month	Signal	z	Direction
IMIP Morowali	Sep 2026	surface change	+4.4	up
Morenci	Sep 2026	pond / water area	+4.0	up
IMIP Morowali	Sep 2026	tree cover	-3.8	down
IWIP Weda Bay	Sep 2026	footprint	+3.6	up
IMIP Morowali	Sep 2026	footprint	+3.4	up
Kamoa-Kakula	Sep 2026	tree cover	-3.4	down
IMIP Morowali	Sep 2026	NO2	+3.3	up
IWIP Weda Bay	Sep 2026	night lights	+3.3	up
IMIP Morowali	Sep 2026	night lights (mean)	+3.2	up
Kamoa-Kakula	Sep 2026	night lights	+3.2	up
IMIP Morowali	Sep 2026	night lights	+3.1	up
Salar de Atacama ponds	Sep 2026	pond / water area	+3.1	up
Greenbushes	Sep 2026	bare ground	+2.8	up
Kamoa-Kakula	Sep 2026	footprint	+2.7	up
IMIP Morowali	Sep 2026	SO2	+2.7	up
Kamoa-Kakula	Sep 2026	night lights (mean)	+2.7	up
Kamoa-Kakula	Sep 2026	pond / water area	+2.6	up
IMIP Morowali	Sep 2026	bare ground	+2.6	up

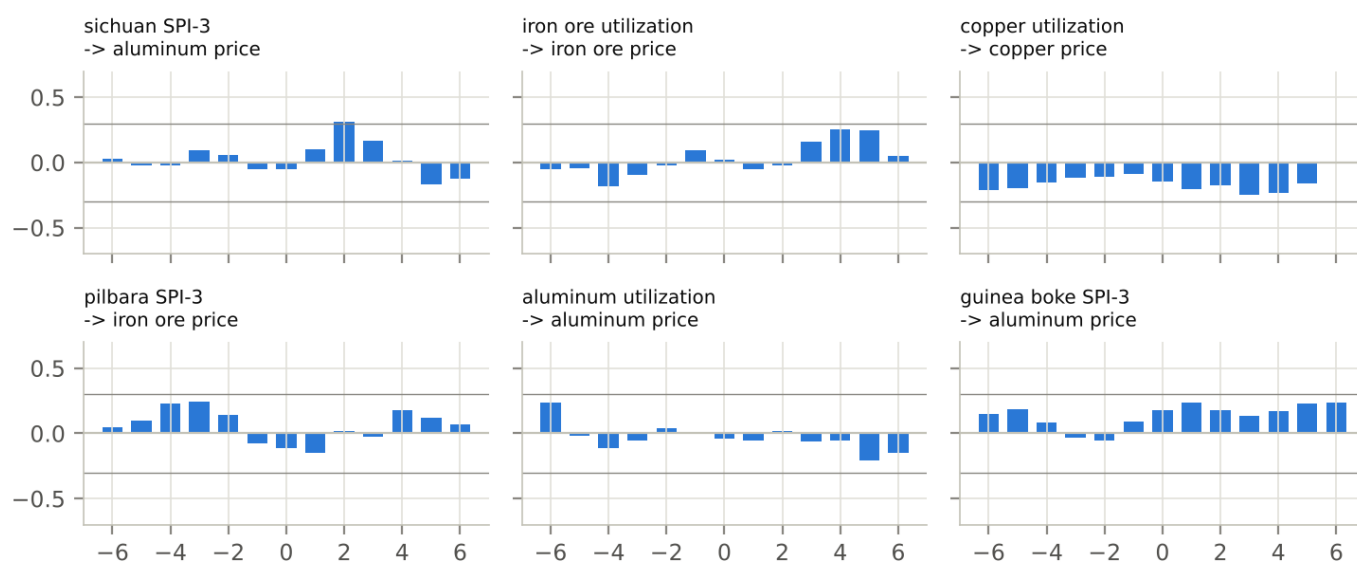
155 more in outputs/anomalies_recent.csv.

Price linkage and validation

Each pair shows the lag with the largest $|r|$ between the signal level and the next months' log returns ($k > 0$ means the satellite signal leads). p uses an autocorrelation-adjusted sample size but not a multiple-testing correction across 13 lags, so read these as hypotheses.

Pair	Best lag	r	n	n_eff	p (approx.)
sichuan SPI-3 -> aluminum price	+2	+0.31	91	86	0.003
iron ore utilization -> iron ore price	+4	+0.25	60	59	0.051
copper utilization -> copper price	+3	-0.25	63	57	0.059
pilbara SPI-3 -> iron ore price	-3	+0.25	88	87	0.019
aluminum utilization -> aluminum price	-6	+0.24	66	65	0.051
guinea boke SPI-3 -> aluminum price	+6	+0.24	87	81	0.030
palladium utilization -> palladium price	+2	+0.20	62	77	0.085
zambezi kariba SPI-3 -> copper price	+6	+0.19	87	74	0.105
nickel utilization -> nickel price	-1	+0.16	65	69	0.196
yunnan SPI-3 -> aluminum price	+6	-0.14	87	82	0.193

Correlation of signal level with monthly log return, top 6 pairs (lines at ± 0.3)



lag k in months ($k > 0$: satellite signal leads price)

Validation against reported production

Site	Frequency	Satellite index	Reported output	r	n
Escondida (BHP)	quarterly	utilization index	production level	-0.12	22
Escondida (BHP)	quarterly	utilization index	production YoY %	-0.12	22
Escondida (BHP)	quarterly	capacity index	production level	-0.37	23
Escondida (BHP)	quarterly	capacity index	production YoY %	-0.29	23

Validation is the real test of a satellite proxy: a useful site index should co-move with reported output before it is trusted for prices.

Data coverage (% of months missing after quality control)

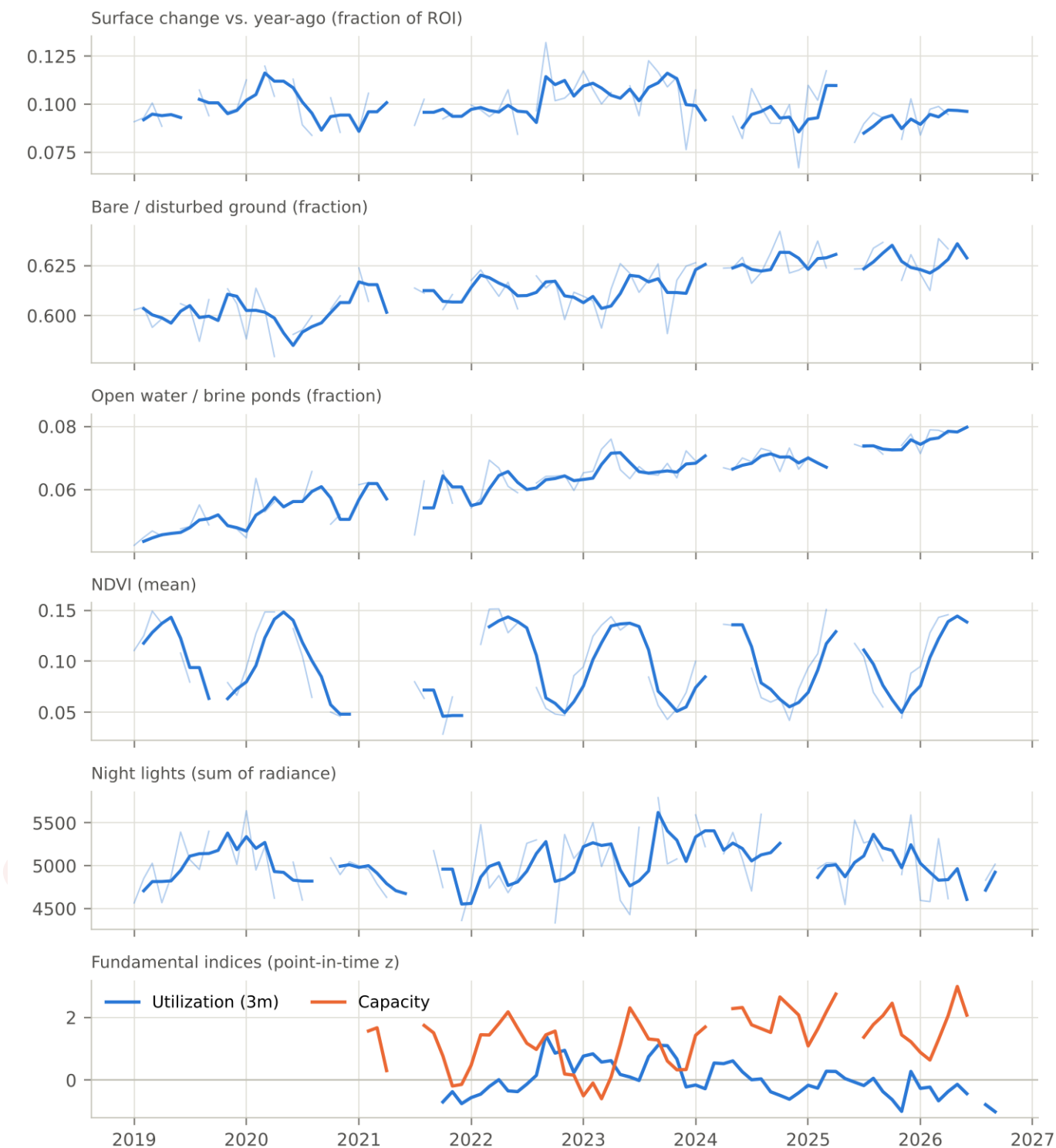
Site	Sentinel-2	Sentinel-1	Sentinel-5P	VIIRS	FIRMS	Dynamic World
Chuquicamata mine + smelter	29%	-	16%	17%	15%	-
Cobre Panama pits + plant	27%	15%	-	22%	-	25%
Cobre Panama Punta Rincon port + coal plant	-	24%	15%	21%	-	-
Escondida	24%	-	-	19%	-	-
Grasberg district	28%	21%	-	21%	-	-
Greenbushes	34%	-	-	19%	-	21%
Kamoa-Kakula	35%	-	15%	20%	18%	23%
Port Kamsar bauxite terminal	27%	18%	-	24%	-	-
Morenci	34%	-	-	10%	-	-
IMIP Morowali	27%	19%	16%	15%	23%	19%
Mt Whaleback / Newman	29%	-	-	21%	-	-
Norilsk complex	-	-	18%	23%	23%	-
Port Hedland export hub	29%	31%	-	22%	-	-
Salar de Atacama ponds	28%	-	-	-	-	-
IWIP Weda Bay	23%	23%	20%	19%	-	22%

- = signal not used at that site. Gaps come from cloud (optical), polar night or summer, sensor outages and the quality rules in docs/methodology.md.

Site appendix: Escondida (BHP)

copper | mine | Chile | ROI -24.270, -69.070, radius 9 km | signals: s2, viirs
World's largest copper mine; hyper-arid, so optical imagery works year-round.

Escondida (BHP) - copper - mine



Site appendix: Chuquicamata mine + smelter (Codelco)

copper | mine smelter | Chile | ROI -22.290, -68.900, radius 10 km | signals: s2, s5p, viirs, firms
Smelter SO2 tracks throughput; Calama town adds NO2 background.

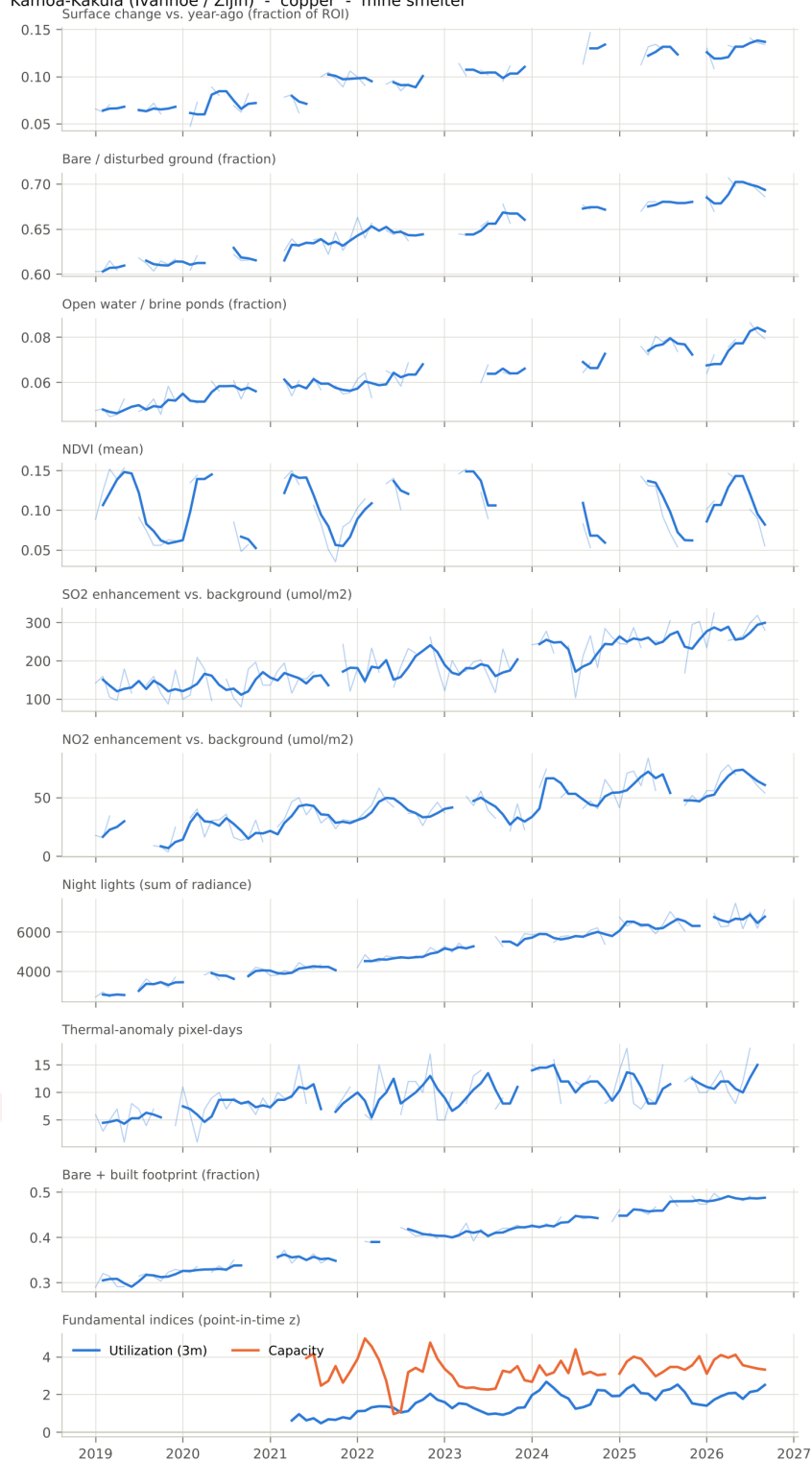
Chuquicamata mine + smelter (Codelco) - copper - mine smelter



Site appendix: Kamo-a-Kakula (Ivanhoe / Zijin)

copper | mine smelter | DRC | ROI -10.887, 25.181, radius 12 km | signals: s2, dw, viirs, s5p, firms
Underground; surface signal comes from concentrators, tailings and the on-site smelter.

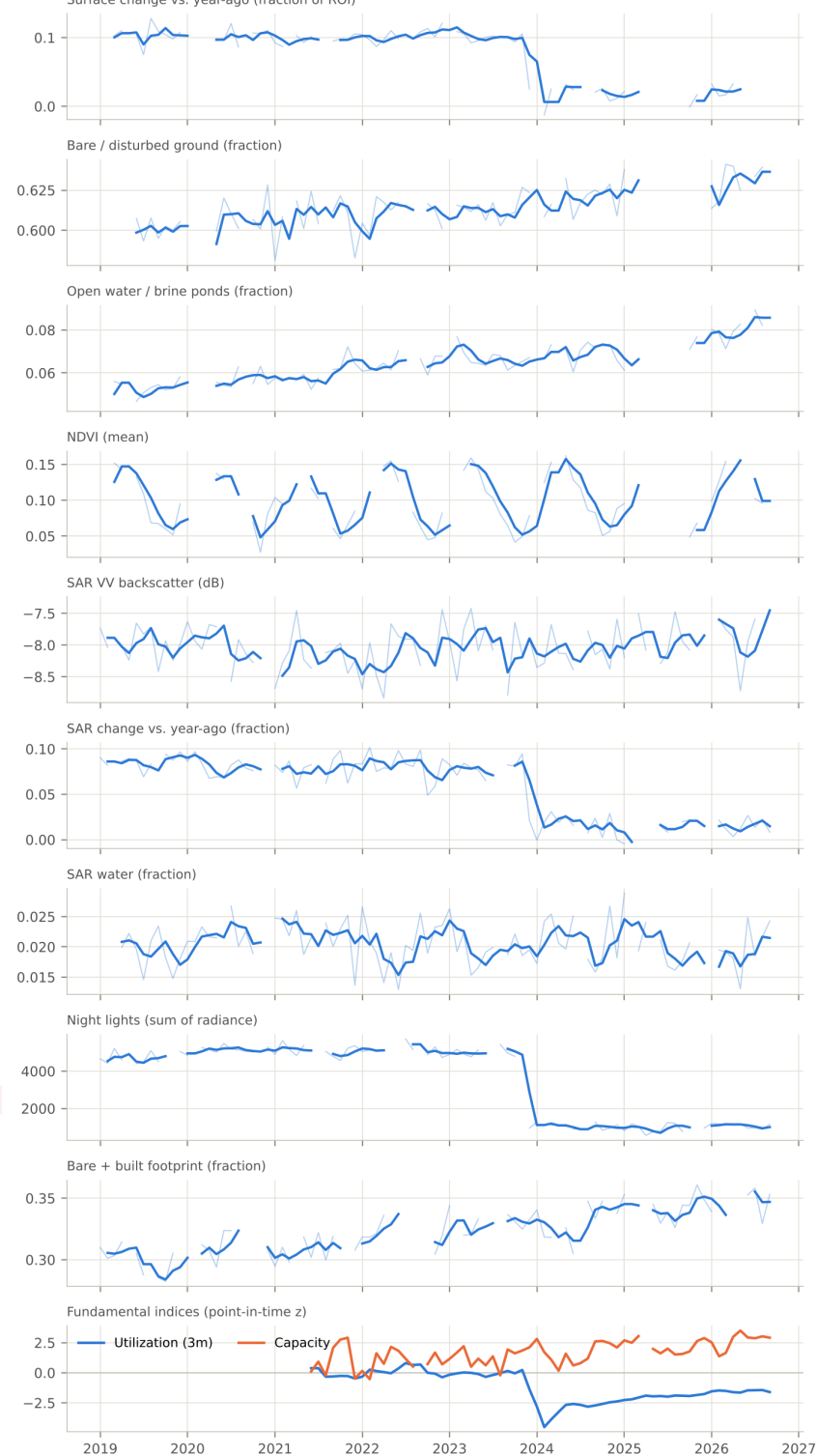
Kamo-a-Kakula (Ivanhoe / Zijin) - copper - mine smelter



Site appendix: Cobre Panama pits + plant (First Quantum)

copper | mine | Panama | ROI 8.840, -80.640, radius 8 km | signals: s1, s2, dw, viirs
Halted late 2023, a natural test case for every signal. Persistent cloud, so lean on SAR.

Cobre Panama pits + plant (First Quantum) - copper - mine
Surface change vs. year-ago (fraction of ROI)

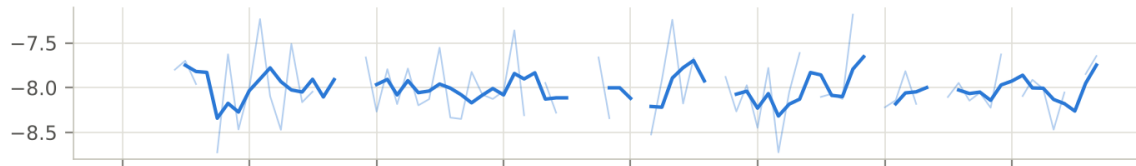


Site appendix: Cobre Panama Punta Rincon port + coal plant

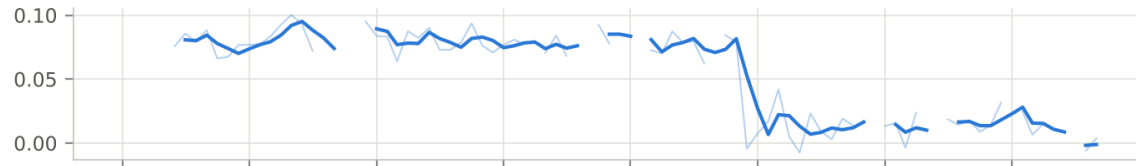
copper | port | Panama | ROI 9.006, -80.690, radius 4 km | signals: s1, viirs, s5p
Captive coal plant NO2 and port lights track operating status.

Cobre Panama Punta Rincon port + coal plant - copper - port

SAR VV backscatter (dB)



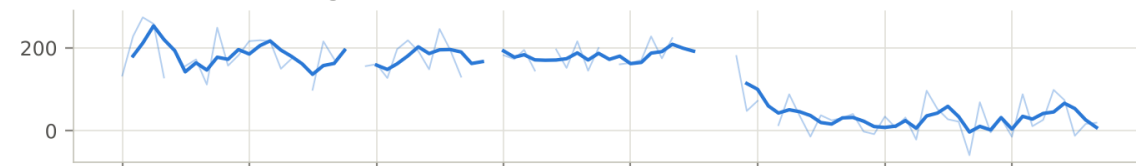
SAR change vs. year-ago (fraction)



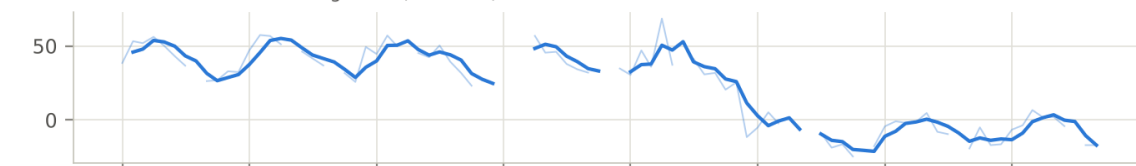
SAR water (fraction)



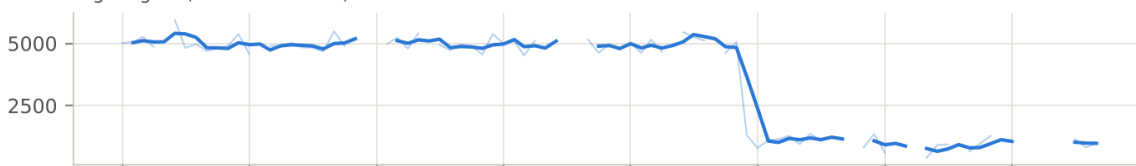
SO2 enhancement vs. background (umol/m2)



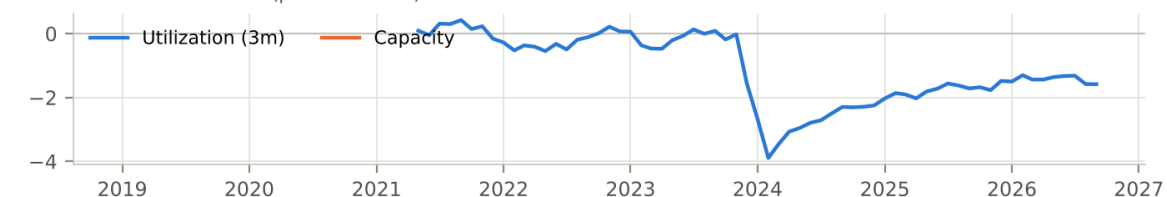
NO2 enhancement vs. background (umol/m2)



Night lights (sum of radiance)



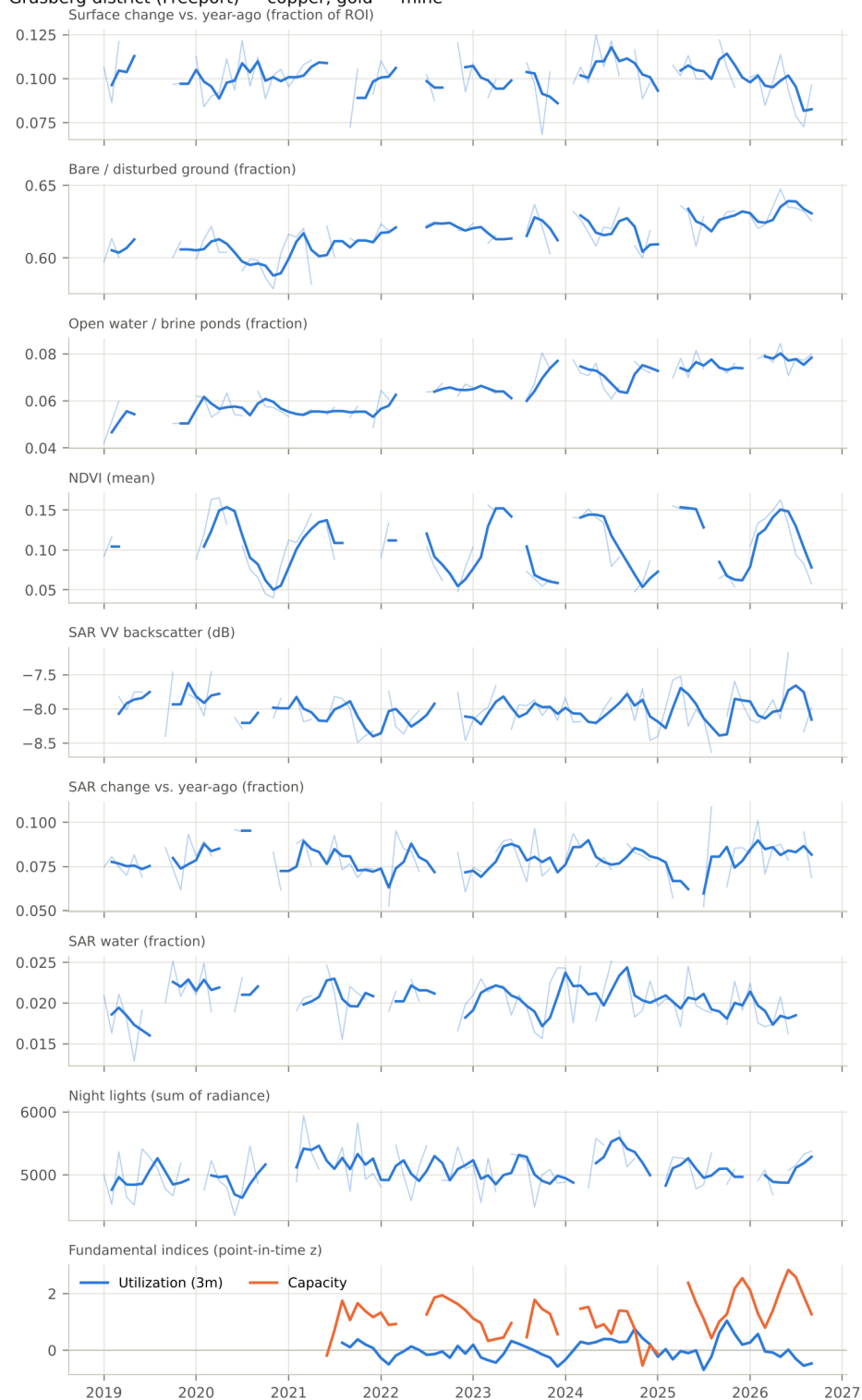
Fundamental indices (point-in-time z)



Site appendix: Grasberg district (Freeport)

copper, gold | mine | Indonesia | ROI -4.050, 137.120, radius 10 km | signals: s1, s2, viirs
Block-cave underground; persistent cloud, so SAR first. Verify the ROI.

Grasberg district (Freeport) - copper, gold - mine



Site appendix: Morenci (Freeport)

copper | mine | USA | ROI 33.080, -109.350, radius 10 km | signals: s2, viirs
Large open pit with heap leach and SX-EW.

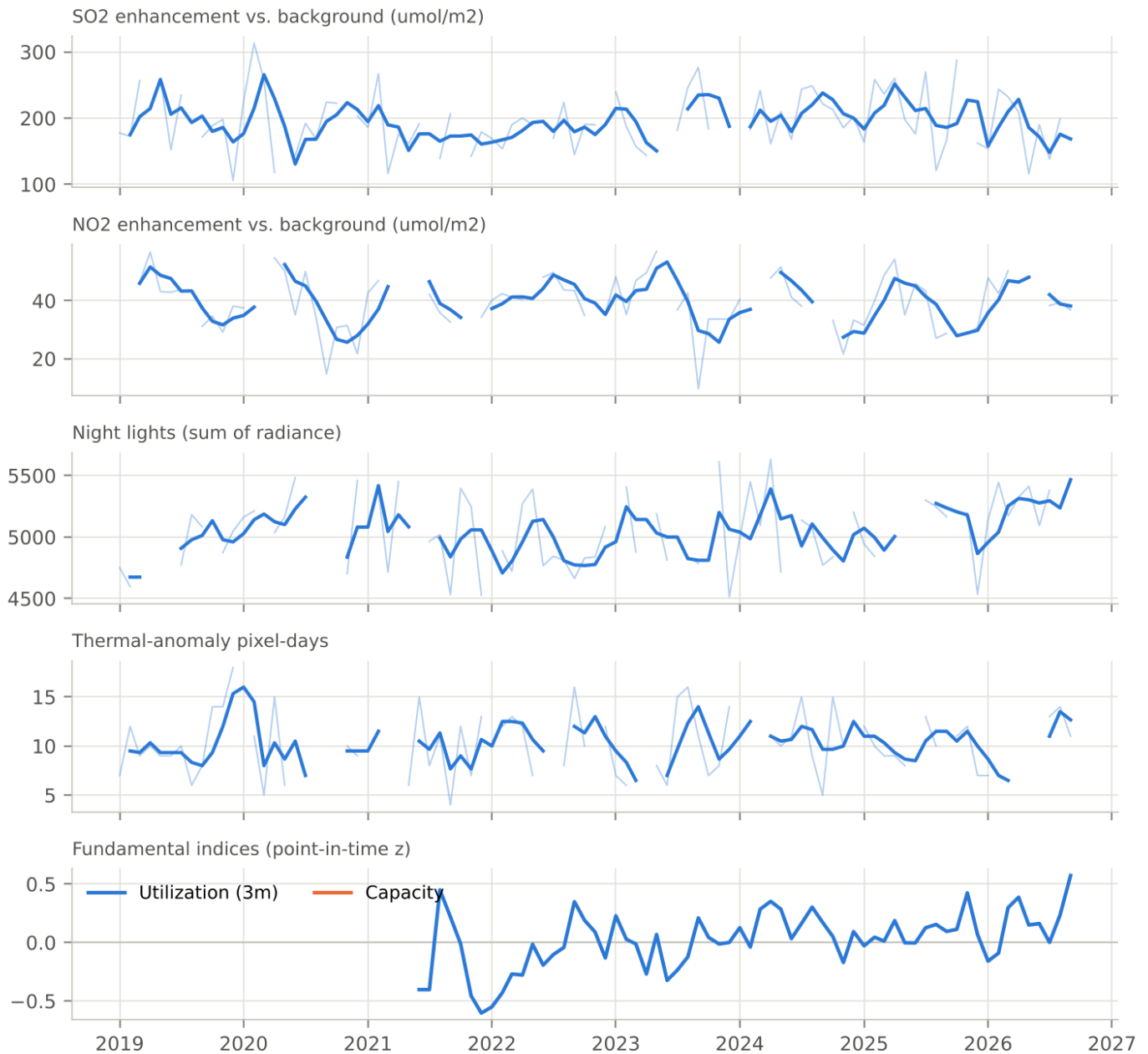
Morenci (Freeport) - copper - mine



Site appendix: Norilsk complex (Nornickel)

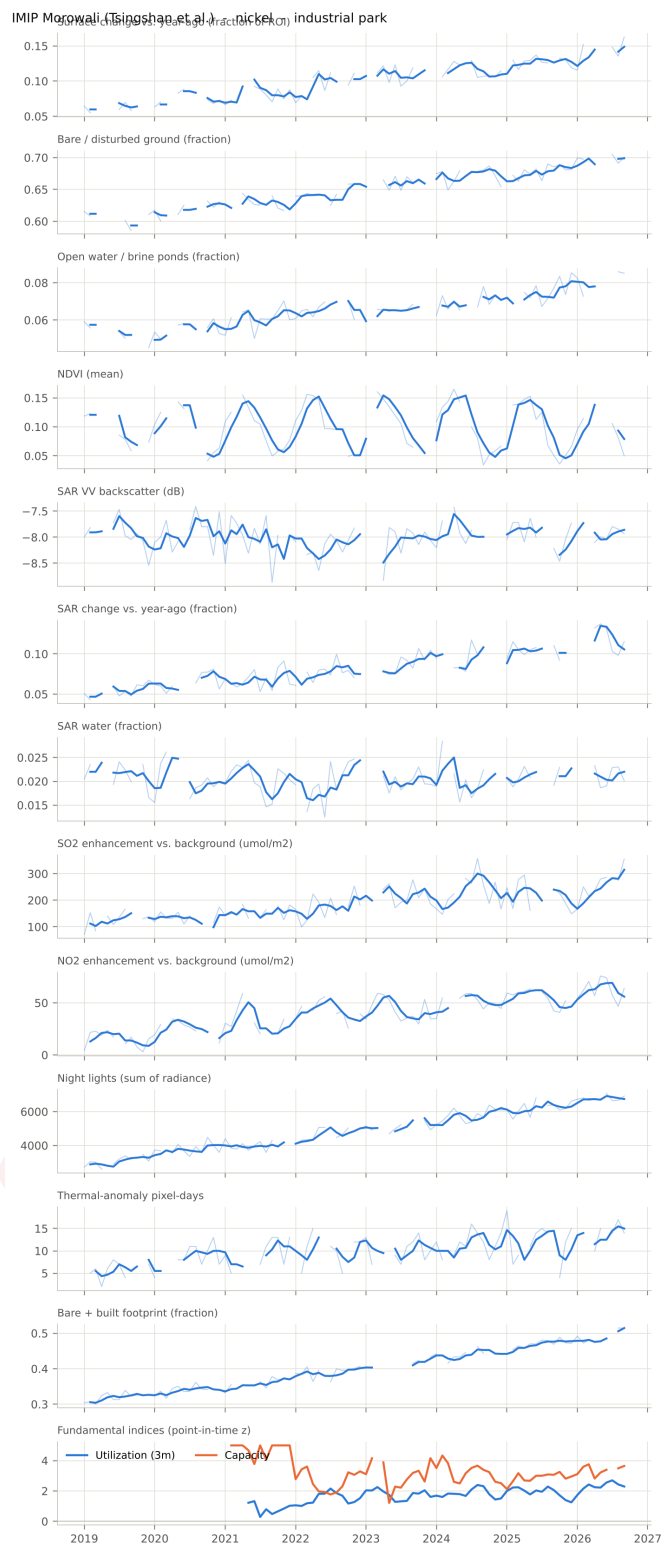
nickel, palladium, copper | smelter | Russia | ROI 69.330, 88.100, radius 30 km | signals: s5p, viirs, firms
Very large point SO₂ source. No S5P in polar night; VIIRS gaps in polar summer.

Norilsk complex (Nornickel) - nickel, palladium, copper - smelter



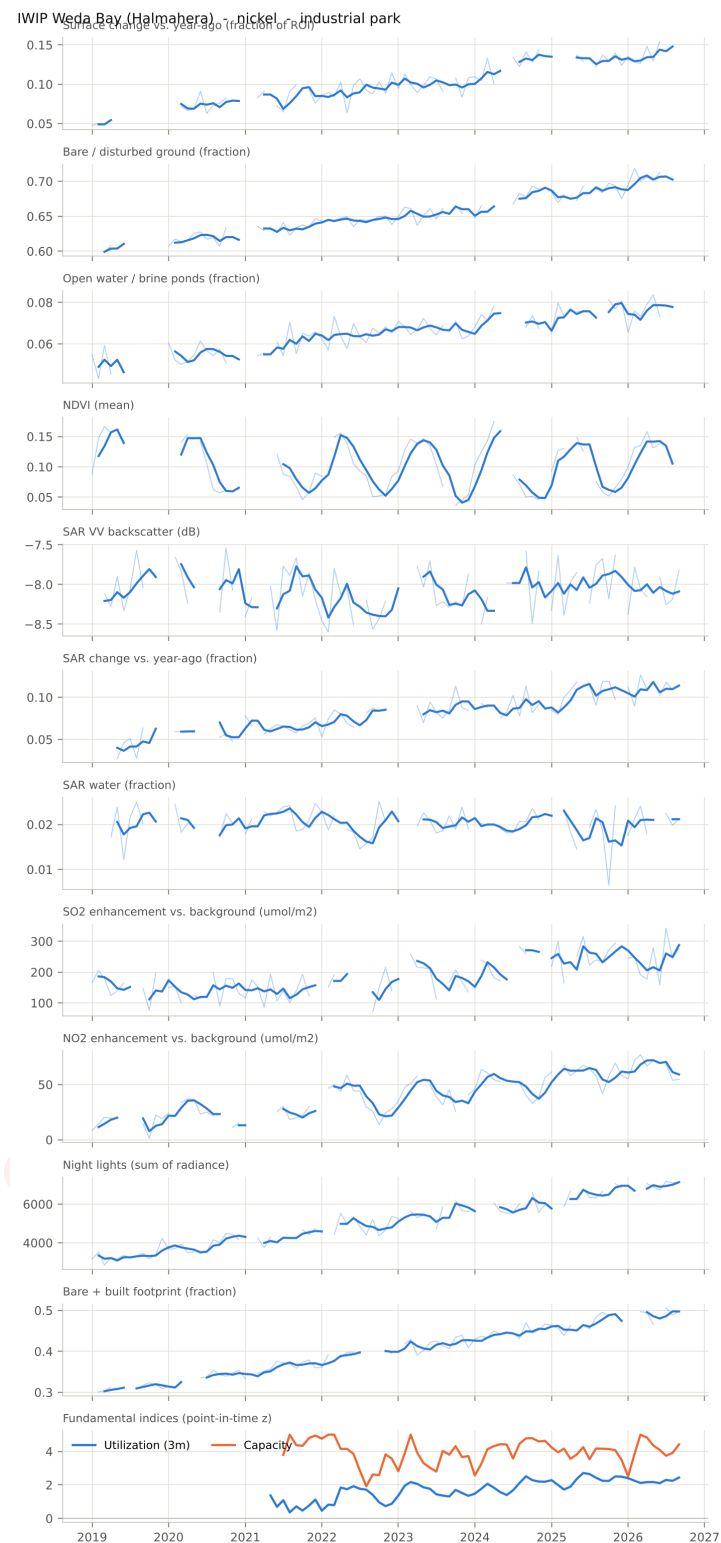
Site appendix: IMIP Morowali (Tsingshan et al.)

nickel | industrial park | Indonesia | ROI -2.830, 122.160, radius 8 km | signals: s1, s2, dw, viirs, s5p, firms
RKEF / NPI and HPAL hub with captive coal power.



Site appendix: IWIP Weda Bay (Halmahera)

nickel | industrial park | Indonesia | ROI 0.476, 128.000, radius 10 km | signals: s1, s2, dw, viirs, s5p
Centred on the IWIP captive power station; check that the ROI covers the smelters.

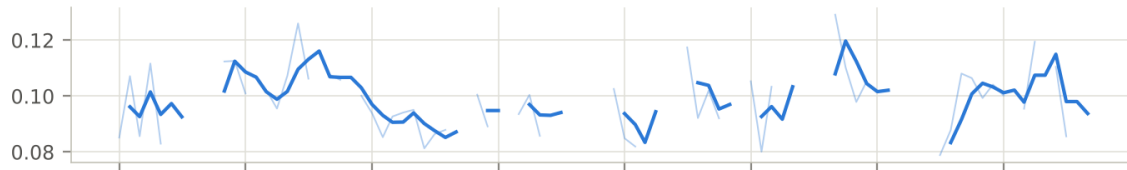


Site appendix: Greenbushes (Talison)

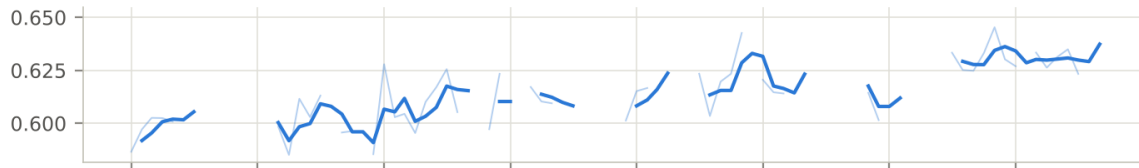
lithium | mine | Australia | ROI -33.860, 116.060, radius 6 km | signals: s2, dw, viirs
Hard-rock spodumene; pit and waste-dump growth is the capacity proxy.

Greenbushes (Talison) - lithium - mine

Surface change vs. year-ago (fraction of ROI)



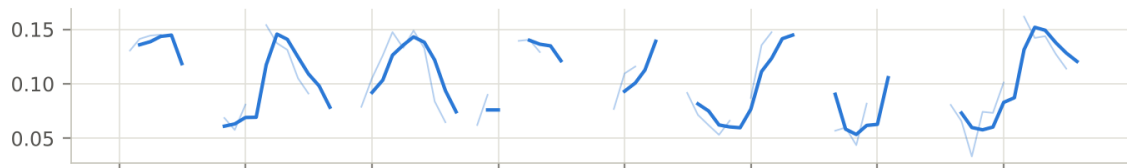
Bare / disturbed ground (fraction)



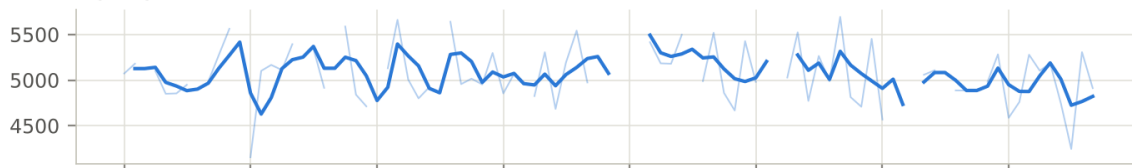
Open water / brine ponds (fraction)



NDVI (mean)



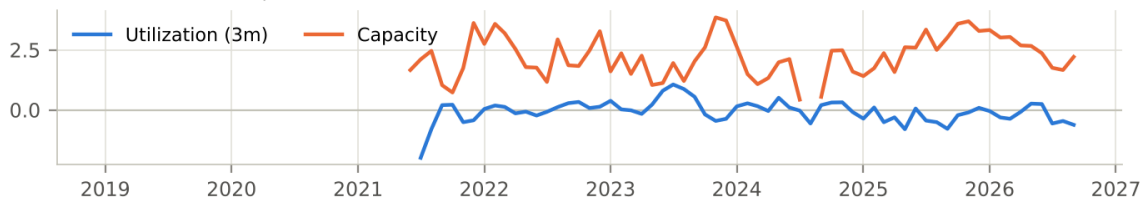
Night lights (sum of radiance)



Bare + built footprint (fraction)



Fundamental indices (point-in-time z)



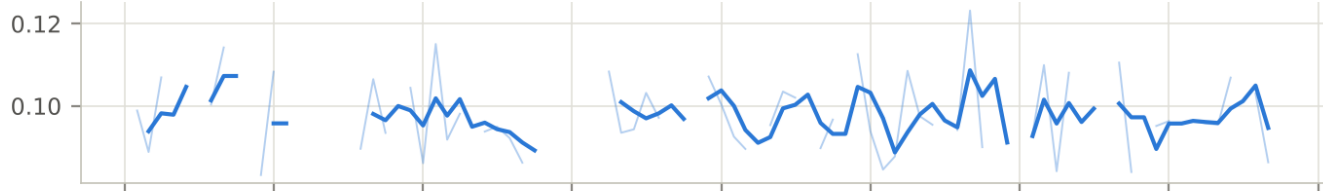
Site appendix: Salar de Atacama ponds (SQM / Albemarle)

lithium | brine | Chile | ROI -23.620, -68.360, radius 18 km | signals: s2

Pond area (MNDWI water fraction) tracks brine in evaporation. Approximate - verify the ROI.

Salar de Atacama ponds (SQM / Albemarle) - lithium - brine

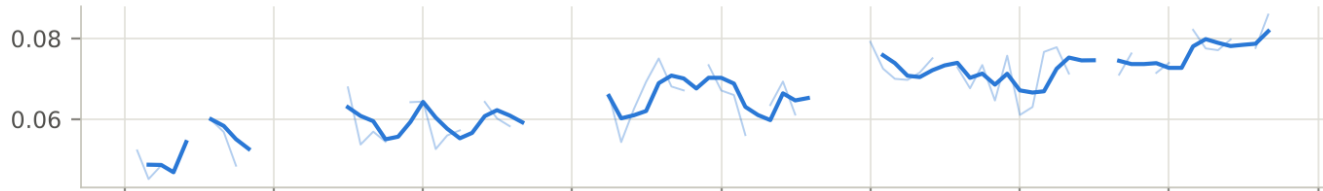
Surface change vs. year-ago (fraction of ROI)



Bare / disturbed ground (fraction)



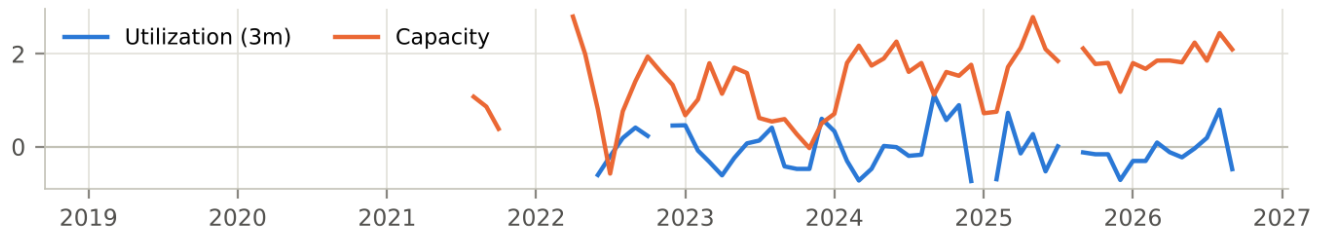
Open water / brine ponds (fraction)



NDVI (mean)



Fundamental indices (point-in-time z)

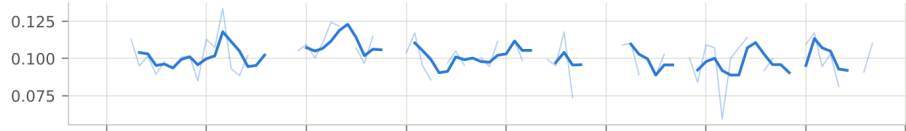


Site appendix: Port Hedland export hub

iron_ore | port | Australia | ROI -20.310, 118.580, radius 8 km | signals: s1, s2, viirs
Stockpile yards and berths; SAR change and night lights track throughput.

Port Hedland export hub - iron_ore - port

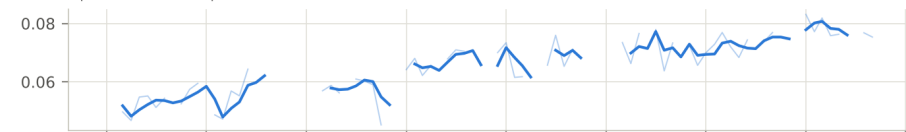
Surface change vs. year-ago (fraction of ROI)



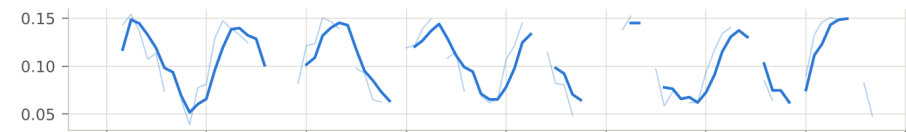
Bare / disturbed ground (fraction)



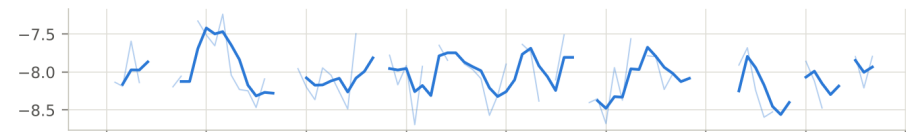
Open water / brine ponds (fraction)



NDVI (mean)



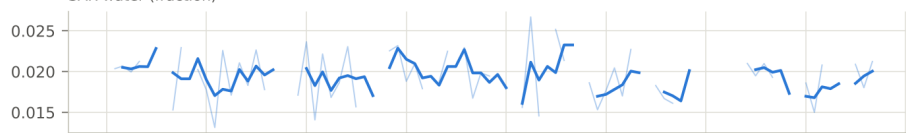
SAR VV backscatter (dB)



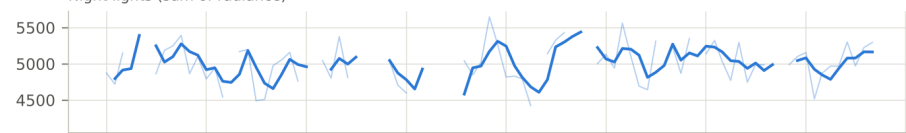
SAR change vs. year-ago (fraction)



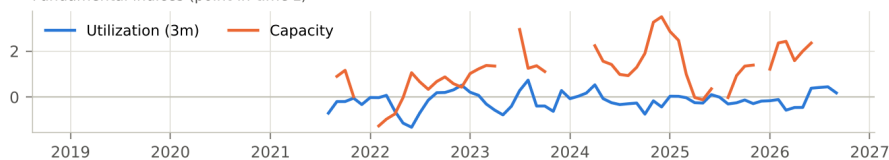
SAR water (fraction)



Night lights (sum of radiance)



Fundamental indices (point-in-time z)

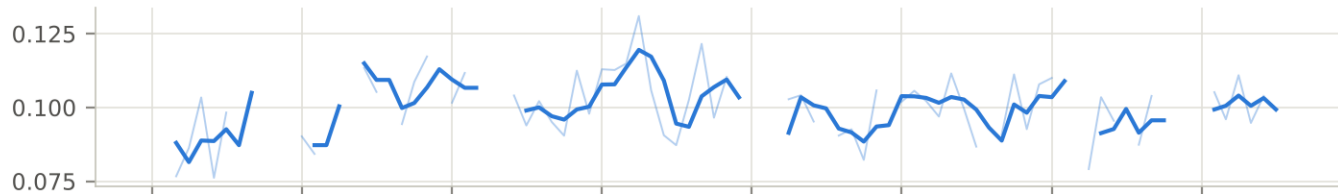


Site appendix: Mt Whaleback / Newman (BHP)

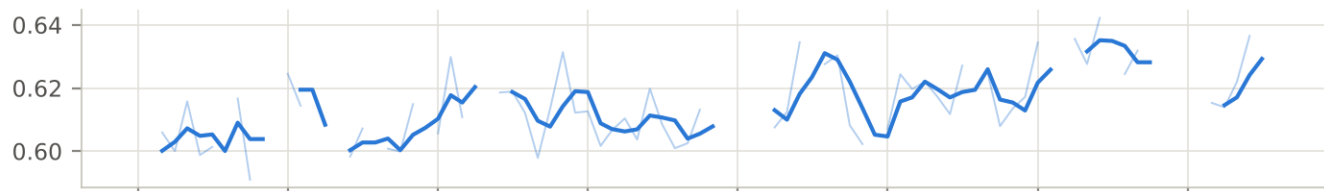
iron_ore | mine | Australia | ROI -23.360, 119.670, radius 8 km | signals: s2, viirs

Mt Whaleback / Newman (BHP) - iron_ore - mine

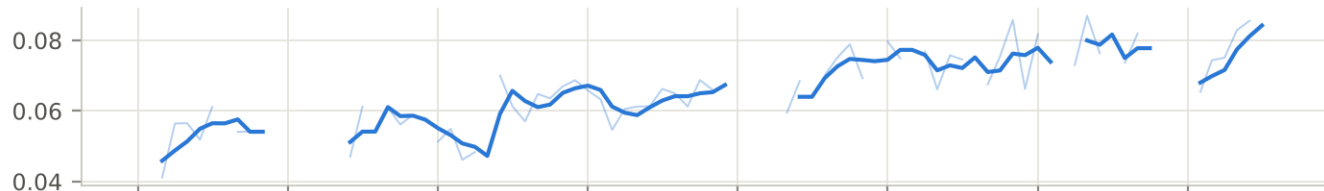
Surface change vs. year-ago (fraction of ROI)



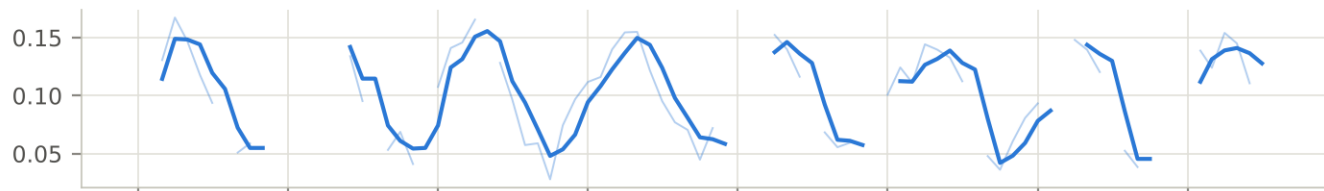
Bare / disturbed ground (fraction)



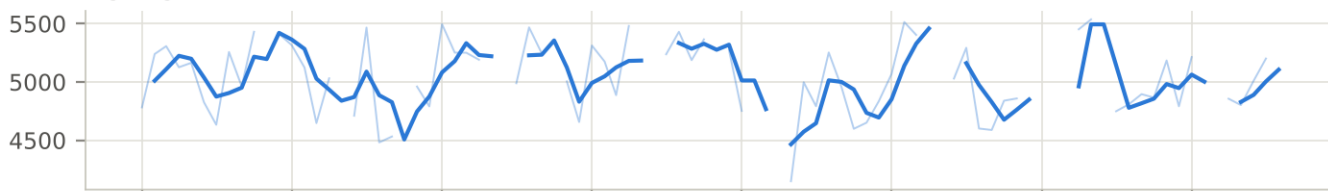
Open water / brine ponds (fraction)



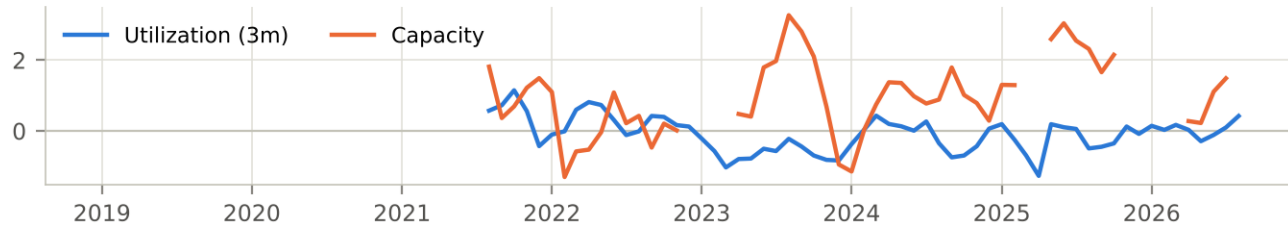
NDVI (mean)



Night lights (sum of radiance)



Fundamental indices (point-in-time z)

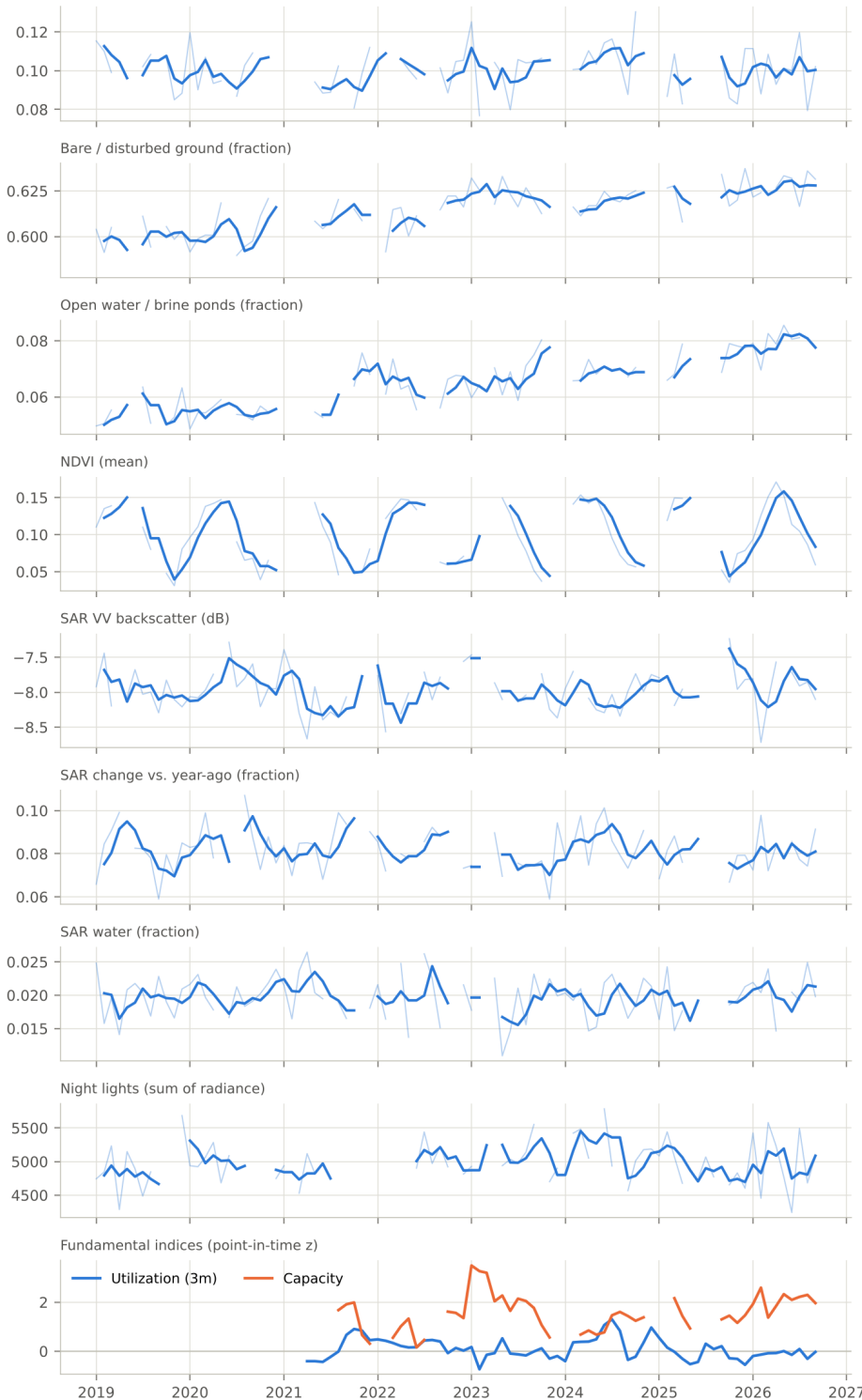


Site appendix: Port Kamsar bauxite terminal (CBG)

bauxite, aluminum | port | Guinea | ROI 10.650, -14.610, radius 6 km | signals: s1, s2, viirs
Wet season (about Jun-Oct) slows loading; SAR sees through cloud.

Port Kamsar bauxite terminal (CBG) - bauxite, aluminum - port

Surface change vs. year-ago (fraction of ROI)



Methodology and caveats

What is measured

Each site is a circle (region of interest) around a mine, smelter, industrial park, port or brine field. Every month the pipeline reduces satellite data over it in Google Earth Engine: Sentinel-2 surface reflectance with Cloud Score+ masking (bare-soil index, NDVI, MNDWI water/pond fraction and the share of pixels that changed against the same month a year earlier); Sentinel-1 SAR in one orbit direction (backscatter, SAR water, year-on-year change); Sentinel-5P tropospheric NO₂ and SO₂ as the site mean minus a background ring 40-90 km out; VIIRS monthly night lights screened by cloud-free coverage; MODIS FIRMS thermal-anomaly pixel-days; and the Dynamic World bare-plus-built footprint. Supply regions get CHIRPS rainfall and ERA5-Land temperature.

How signals become indices

Each signal is converted to a point-in-time seasonal z-score: the value minus the mean of the same calendar month in prior years, divided by the standard deviation of past anomalies. Only data available at that month is used, so the series can be backtested without look-ahead. The utilization index averages the signed z-scores of throughput proxies; the capacity index averages z-scores of 3-month-smoothed footprint proxies. Commodity indices are weighted means across sites (weights in config/sites.yaml, default equal). Rainfall risk uses an SPI-3-like standardised 3-month total against a 2001-2020 baseline by calendar month; flags fire on the side of the distribution that hurts supply for that region (drought for hydro-powered smelting, extreme rain for haulage and ports).

How to read it

These are activity proxies, not production measurements. A z-score near +/-2 says a signal is unusual for that site and month; it does not say why. Confirm moves against company reports, customs and trade data, or news before acting. Cloud cover (tropics), polar night (no Sentinel-5P), polar summer (no night lights) and sensor gaps blank months, which the coverage table shows. Underground mines expose only their surface footprint. Price correlations on fewer than ~90 monthly points are fragile, and the best of 13 lags will look significant by chance more often than its p-value suggests.

Site locations

Coordinates are approximate centroids. Each live run should be preceded by `bash run.sh --check-sites`, which saves a cloud-free image of every site with its outline for visual confirmation.

Data sources and attribution

Contains modified Copernicus Sentinel data (Sentinel-1, Sentinel-2, Sentinel-5P), processed in Google Earth Engine. Cloud Score+ S2_HARMONIZED V1 (Google, CC-BY-4.0). Dynamic World V1 (Google and World Resources Institute, CC-BY-4.0). VIIRS Day/Night Band monthly composites, Earth Observation Group, Payne Institute for Public Policy, Colorado School of Mines (public domain). FIRMS MODIS active fire / thermal anomalies, NASA LANCE / EOSDIS. Near-real-time data are not science quality; see the LANCE disclaimer at earthdata.nasa.gov. CHIRPS rainfall, UC Santa Barbara Climate Hazards Center. ERA5-Land monthly aggregates, ECMWF / Copernicus Climate Change Service. Prices: CME Group futures via Yahoo Finance (yfinance), for research use.